D06 · Carrier Management & TMS

03 · Route optimization & territory design

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1 WHAT IT IS

Route optimization assigns stops to vehicles and sequences them to minimize distance, time or cost under real constraints — time windows, capacity, driver hours, vehicle restrictions. Territory design divides the service area into balanced zones that make daily routing stable.

2 WHY IT MATTERS

The vehicle routing problem is one of the most studied problems in operations research, and modern solvers handle it well **(Toth and Vigo — Vehicle Routing, SIAM)**. **The practical gap is not the algorithm; it is the constraints.** Routes that ignore circulation restrictions, unloading windows or security conditions in cities like CDMX, Guadalajara or Monterrey will be rejected by drivers and dispatchers, however optimal they look.

3 HOW IT IS DONE

1. **Design territories before routes.** Balance workload, geography and service requirements so the daily optimizer starts from a sound structure.
2. **Model the real constraints.** Time windows, truck bans, height and weight limits, driver hours and security rules belong in the model.
3. **Combine fixed and dynamic routing.** Fixed backbone routes for stable high-volume customers, dynamic optimization for the variable tail.
4. **Close the loop with actuals.** Compare planned vs. driven routes, feed back service times and re-tune parameters; review territories at least annually.

IN PRACTICE

Redesign territories before optimizing daily routes

Daily optimization cannot fully compensate for poorly balanced territories.

Model real Mexican urban constraints explicitly

Time windows, truck bans, height restrictions and security considerations must be in the model or the optimized routes will be rejected by operations.

Review territory design at least annually

Customer density and order patterns drift. Static territories become inefficient within 12–18 months in growing markets.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Distribution company in central Mexico — Fixed routes designed three years earlier no longer matched demand density.

Metric	Before	After territory redesign + dynamic routing
Average stops per route	14.2	18.7
Km per delivery	18.4 km	13.1 km (-29%)
On-time delivery	91%	96%
Fleet utilization	71%	86%

Result: Two vehicles removed from the fleet while improving service. Annual savings approximately $1.1M in operating cost.

HOW IT IS MEASURED

Km per Delivery / Cost per Stop

Total km or total route cost / Number of deliveries

Benchmark: Declining trend is the goal; absolute number depends on geography and density

Stops per Route / Route Utilization %

Average number of productive stops and % of available capacity used

Benchmark: Higher is better until service or driver hours constraints are hit

WHAT YOU WOULD USE

Route Optimization

SAP TM / Blue Yonder / Ortec — Mature VRP engines with support for time windows, vehicle constraints and multi-day planning.

Google OR-Tools / specialized last-mile solvers — Used by companies that build custom optimization layers on top of their TMS.

4 SO WHAT

Optimize inside reality. Fix territories first, model the real urban constraints, mix fixed and dynamic routing, and keep tuning the model with what drivers actually do.

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